For the exclusive use of M. Bagadia, 2026.

Harvard Business School

9-726-358
REV: OCTOBER 23, 2025

DAVID B. YOFFIE
SARAH WALKER

The Voice Wars Continues: Alexa vs. Hey Google vs. Siri vs. ChatGPT in 2025

Amazon CEO Andy Jassy wanted to drive an Alexa comeback in 2025. After a decade of heavy investment in smart speakers and mounting losses, Amazon announced Alexa+, a new version of Alexa powered by Generative AI (GenAI). Alexa+ would be capable of tasks its predecessor could not perform, such as "plan a date, book a restaurant, and text the babysitter. It can create a travel itinerary and add it to everyone's calendar, read your study guide and test you on the answers, remember your favorite movies and food, as well as find your smart devices and take action."[1]

However, Amazon was not the first or the only company to introduce a voice assistant powered with GenAI; OpenAI's ChatGPT announced its own voice assistant in 2024; Apple also introduced GenAI (which it called Apple Intelligence) into Siri's functionality. Google was pushing its voice efforts with its GenAI model called Gemini. Meta was also entering the voice game with its LLAMA open-source models, a GenAI-powered voice assistant app called Meta AI, and GenAI glasses. In 2025, all the voice assistants were imperfect, but voice capabilities were widely viewed as an important user interface for the coming wave of GenAI.

With so much competition, Jassy pondered how Amazon was going to win with voice, and make the economics work. While he noted that Alexa had "over 600 million devices in customers' homes and offices," he also inherited a business unit deeply in the red when he took over from Jeff Bezos in 2021.[2] Alexa had already lost $10 billion in its lifetime.[3] (see **Exhibit 1**) The introduction of GenAI, however, created an opportunity to make a new Alexa, one that was smarter, more powerful, and personalized. As other brands rushed to integrate GenAI into voice assistants, Jassy recognized the need to make swift decisions. The challenge was that GenAI voice assistants remained flawed and much more costly. For Amazon, this posed serious questions: Facing much higher costs, should it charge a subscription fee for its previously free service? Should it stick with smart speakers, where it was dominant, or move aggressively to smartphones, cars, and other embedded applications? Should it rely on advertising, which consumers historically disliked for voice assistants? Or perhaps it should slow-roll its GenAI voice strategy and wait until the technology improves and costs decrease?

Voice Assistants. Voice assistant technology existed since the 1950s, when Bell Laboratories developed the first voice recognition system. Named Audrey, it was designed for hands-free phone dialing in non-consumer settings.[4] By the 1990s, significant advancements had been made in speech

Professor David B. Yoffie and Research Associate Sarah Walker prepared this case. This case was developed from published sources. Funding for the development of this case was provided by Harvard Business School and not by the company. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

Copyright © 2025 President and Fellows of Harvard College. To order copies or request permission to reproduce materials, call 1-800-545-7685, write Harvard Business School Publishing, Boston, MA 02163, or go to www.hbsp.harvard.edu. This publication may not be digitized, photocopied, or otherwise reproduced, posted, or transmitted, without the permission of Harvard Business School.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

recognition. Dragon Systems introduced "Dragon NaturallySpeaking," one of the first speech recognition software programs that could recognize and transcribe spoken words without pauses.[5] Until the early 2010s, voice recognition software remained a high-priced B2B solution. When Apple introduced Siri in 2011, voice assistants entered the B2C space.

To perform a task, most voice assistants would hear a trigger word, record the words in the user's request, and then send the request to a cloud server (since processing requests required significant computational power). The data center would then break down the user's words into individual sounds using algorithms, which would match the individual sounds to other sounds stored in a database. The software would identify the keywords to perform the task, send the information back to the phone or smart speaker, and give the user the requested information.[6]

Before GenAI was introduced, voice assistants relied primarily on Automated Speech Recognition (ASR) and Natural Language Processing (NLP). ASR used statistical analysis to create words and sentences based on the cleaned sound snippets sent by the voice assistant to the cloud server. As one analyst explained:

> Sometimes, words have various forms…For example, "dog" and "dogs" will be interpreted as two different things by computers. When NLP encounters problems like this, a process called Lemmatization is performed. Essentially, we take a word and convert it into its most basic form… After getting rid of the irrelevant filler words, NLR [Natural Language Representation] tries to figure out how the words in a sentence relate to each other by building a tree that assigns a single parent word to each word in the sentence… After all that, the process of extracting meaning becomes much easier.[7]

However, before GenAI, voice assistants were severely limited. The primary uses for smart speakers and voice on a phone were checking the weather, conducting local "near me" searches for restaurants, stores, etc.[8] Voice assistants tapped a limited database, often Wikipedia, to provide real-time answers.

By the mid-2020s, Amazon Echo was the most popular smart speaker, with 61% of American adults owning at least one device, followed by Google Home (23%) and Apple HomePod (16%).[9] Amazon also had the highest number of monthly active users (MAUs), with over 71 million in 2025, followed by 37 million for Google and 24 million for other brands.[10] (see **Exhibits 2**, **3,** and **4**) However, Amazon's lead was dwindling. Smartphones were increasingly using voice commands. In addition, the introduction of GenAI changed the paradigm. Voice assistants such as Alexa and Siri did not support back-and-forth interactions. With GenAI, users suddenly could have multi-turn conversations. GenAI not only drew on a much larger corpus of data, but it could also converse with the user.

By 2024, Google Assistant (which utilized its GenAI model called Gemini) had 92 million voice assistant users in the U.S., followed by Apple's Siri at 87 million users, and Amazon Alexa at 78 million.[11] Unlike Alexa, Google and Apple controlled smartphones in their device ecosystem, allowing them to integrate their voice assistants and reach millions of new customers. One analyst estimated that by 2028, 171 million people would own at least one voice assistant in the United States, with many consumers owning more than one brand and device.[12]

There were generational differences in the adoption of voice assistants. Gen Z (born roughly between 1997 and 2010) grew up using digital technologies. One study suggested that "the most appealing aspects of voice AI for Gen Z are not only the ability to do tasks speedily and hands-free but also the human experience of communicating with their devices."[13] By 2025, almost 95% of Gen Z used voice assistants on their smartphones, while 66% used a smart speaker.[14] For Baby Boomers, voice assistants provided accessibility and safety functions to seniors, as well as helping them fight

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

loneliness.[15] This, along with the aging population in the U.S., also helped drive demand for voice assistants, particularly smart assistants that could perform more complex tasks.

Amazon Echo and Alexa

Jeff Bezos, founder and CEO of Amazon from 1994 to 2021, was one of the early visionaries to see the potential for emerging voice technology. Although Bezos began Amazon as a simple bookstore, he quickly expanded Amazon into one of the world's largest retailers, the world leader in cloud computing with Amazon Web Services (AWS), and an innovator in an array of new tech products, including the Kindle, Fire TV Stick, and a product which became synonymous with Amazon: its Alexa voice assistant platform launched in 2014. Over the next decade, Amazon aggressively pushed smart speakers, often selling at a loss to build an installed base. The Echo and Echo Dot were compact and affordable personal speakers that could perform a variety of tasks, such as playing music or ordering products on Amazon based on a user's purchase history.[16]

The original design for Alexa, named after the libraries at Alexandria, was created by Bezos in 2011 on a conference room whiteboard. He originally planned for the device to be controlled exclusively by a user's voice, with a price of $20. Alexa's virtual control center would be based in the cloud on AWS, letting Amazon update the software without requiring constant hardware upgrades. Alexa's original design contained a speaker, microphone, and mute button, and required Wi-Fi.[17]

Bezos hoped to make Alexa the first popular voice assistant. However, in October 2011, Apple launched Siri, which was completed shortly before Steve Jobs' passing.[18] Although the Alexa team was discouraged, they saw the potential to differentiate Alexa from other voice assistants. Unlike Siri, Alexa did not need a smartphone to operate, and the team was developing a technology that could understand voices "across a noisy room."[19] Within two years, Amazon had also acquired several startups to accelerate development,[20] including Evi, a little-known British company.[21] Initially, Amazon planned to leverage Evi's technology to build an artificial speech-based book reader.[22] This narrow vision later evolved into an idea to create a new intelligent assistant platform powered by a combination of AWS, speech recognition, and high-quality speech synthesis, which would be tied to affordable dedicated hardware. The idea became a reality in late 2014 when Amazon offered the Alexa-powered Echo smart speaker to select Prime members for $99.[23]

In 2015, Amazon announced the Alexa Skills Kit, a collection of self-service application programming interfaces (APIs) and tools that made it easy for third-party developers to create new Alexa skills. Skills, in effect, were the equivalent of third-party applications. By 2022, Alexa had over 10,000 people working on the product at Amazon, 50,000 third-party developers, and 124,000 Alexa skills, far more than its rivals.[24] However, the growth of new skills fell rapidly in the 2020s.[25] The vast majority of available skills were never used. By 2025, Amazon had sold around 600 million Alexa devices.[26] While smart speaker sales continued to increase, growth was slowing.[27]

Selling many devices did not lead to profitability. Most of the skills used on Alexa were not revenue-generating; hardware sales were low-margin, and e-commerce sales through Alexa were disappointing. (see **Exhibit 5**) Amazon leadership had hoped that Echo devices would boost retail sales, but many owners did not use them for shopping. A 2018 survey conducted by The Information found that only 2% of U.S. consumers used the voice feature to make a purchase, and of these 2%, 90% reported that they would not use it again for shopping.[28] By late 2024, consumer attitudes towards ecommerce in voice assistants appeared to be shifting. A PwC survey stated that 50% of consumers had made a purchase using voice assistants, with another 25% stating they would possibly in the future.[29] This caused an increase in the use of voice assistants to access internet search engines. One

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

study noted that 71% of people preferred voice search over traditional typing "because it provides them with easier and swifter information access."[30]

New leadership, New Strategy In July 2021, Andy Jassy was named Amazon's new CEO. Bezos stayed on as Executive Chairman.[31] After graduating from Harvard Business School in 1997, Jassy went to work for Amazon. By 2003, he had become the leader of the AWS team. Jassy was officially named the CEO of AWS in 2016, and by 2020, AWS managed approximately 1/3 of the total cloud computing market. Upon taking Amazon's helm, Jassy was confronted with several problems that required immediate attention. After over-hiring and over-expanding its distribution centers during the COVID pandemic, Amazon's sales dropped significantly. By 2022, Amazon was the first company to lose $1 trillion in market value.[32] One of Jassy's immediate challenges was to scale back investments made by Bezos. In 2022, Amazon laid off 10,000 employees, with the Alexa division being one of the hardest hit. After almost $3 billion in losses just in the first quarter of 2022, Alexa had gone from the golden project of Bezos to the "swift downfall of the voice assistant and Amazon's larger hardware division."[33] As one analyst stated, "Jassy was handed a deck of cards that had a lot of cleanup required."[34]

Jassy implemented changes across Amazon's policies and procedures to reverse the losses. He prioritized frugal measures, stating, "Think about, if this were my money, would this be the way I would spend it?" One of the most notable changes Jassy implemented was killing "moonshot" projects, which were time-consuming and expensive. By 2023, he had cut 27,000 jobs. However, Jassy received pushback on these measures from employees. They were concerned that Amazon lacked the mindset that led to innovative products like Alexa in the first place.[35]

Nonetheless, Jassy's cost-cutting produced immediate results. In 2023, Amazon's operating income tripled to $15.3 billion, the largest quarterly profit since the company's founding. Growth also accelerated, and in 2024, Amazon's market cap reached $2 trillion.[36] However, while AWS was benefiting from the GenAI revolution, Amazon's products and services were falling behind in the GenAI race. By mid-2024, Amazon had "all-hands-on-deck" to prepare multiple GenAI projects.[37] One of the main focuses was on integrating GenAI into Alexa. According to several employees, Jassy was taking a more methodical approach to investing in AI compared to Bezos. As one person explained, "Alexa was chaotic. Gen AI is not like that. We're not just throwing things at the wall."[38]

2025: Amazon Releases Alexa+ In March 2025, Amazon announced Alexa+, the GenAI version of Alexa. With expanded capabilities, Amazon planned to broaden Alexa+'s appeal and audience. As Panos Panay, Amazon's head of Devices and Services, explained, "Alexa did need to get better. It did need to grow. It did need to get to the next level…Our goal wasn't just to have a chatbot; it was to have Alexa be able to get stuff done."[39] Amazon planned to release new functionalities in layers over time, announcing that some of the functions were "in development and will be released in future updates."[40] Amazon announced that Alexa+ would be $19.99 per month as a standalone service, or a free option for all Prime users. At the time, the individual Prime membership was $14.99 per month.[41]

Alexa+ used foundational models to deliver new capabilities as well as deliver "classic" Alexa tasks. For example, unlike the classic version of Alexa, users would not need to repeatedly use the word "Alexa" to activate a session, since Alexa+ would automatically understand that the user was talking to it through context clues and memorized personal information about the user. LLMs powered Alexa+ from Anthropic and one of Amazon's internal models called Nova.[42] According to Amazon, Nova was "a new generation of state-of-the-art foundation models (FMs) that deliver frontier intelligence and industry-leading price performance."[43] In March 2025, Amazon released the Nova Act and the Nova Act SDK and toolkit. Nova Act was a "general-purpose AI agent" that could perform simple tasks, such as controlling a web browser. An SDK was designed for developers to create tools that automate

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

basic functions for their users, such as ordering food or making dinner reservations.[44] As of July 2025, Alexa+ was still in the testing phase but had rolled out to 1 million users, according to Amazon.[45] For comparison, Amazon Prime had around 200 million members.

The features known as "Skills" in the original Alexa were renamed "Experts" for Alexa+. Experts utilized numerous LLMs to process user queries in the cloud. Amazon stated it had tens of thousands of Experts integrations through APIs and "agentic experiences," allowing Alexa+ to "navigate and fill out web forms on your behalf to do things like book a cleaner or repairman."[46] "Agentic experiences" also meant that Alexa+ would have "autonomous capabilities," allowing it to complete tasks without relying exclusively on human oversight.[47] This was a distinguishing feature of Alexa+ compared to the original Alexa, which required human intervention to complete similar tasks.

Initial reviews of Alexa+ were mixed. Alexa+ excelled at conducting natural-sounding conversations, a feat that was previously impossible for the original Alexa. Alexa+ was also capable of remembering past conversations, had the capability to book a table at a restaurant, and could handle multiple requests at one time.[48] Yet one review asserted that Alexa+ was "two steps forward, one step back."[49] Despite advanced new features, Alexa+ was "too buggy and unreliable…to recommend."[50] It hallucinated, made "inexplicable errors," and had "excruciating" delays in responding to queries.[51] While the original Alexa had been one of the most popular voice assistants in the market, the verdict was still out on whether Alexa+ could compete with other voice assistants for more complex tasks.[52]

Competition in the Voice Assistant Landscape

Google (Alphabet) In 2015, Sundar Pichai was named the new CEO of Google.[53] An early advocate of AI at Google, Pinchai released a company-wide memo in 2016 stating, "…in the next 10 years, we will shift to a world that is AI-first, a world where computing becomes universally available – be it at home, at work, in the car, or on the go..."[54] In response to Amazon's introduction of Alexa, Google introduced its own smart speakers in 2016.[55] The same year, Pichai marketed Google Assistant for Android as a "conversational assistant…We want users to have an ongoing, two-way dialogue with Google. We want to help you get things done in your real world… We think of building each user their own individual Google."[56]

By 2019, Google Assistant was on a billion devices, mostly smartphones. By 2020, Google Assistant was experiencing YoY growth of 17%.[57] Google had developed the Tensor chip to run deep learning models directly on its Pixel smartphones to reduce latency for tasks like transcription and real-time translation. Through their SDK that enabled third-party hardware manufacturers to add Google Assistant to almost any device, manufacturers such as Sony, Panasonic, and Kohler embedded Google Assistant into their smart speakers, TVs, and appliances. By 2018, Google claimed to have over one million actions (essentially instructions) that, among many other things, translated languages, shopped online, and offered games for children.[58] These Actions included an API for 3rd party developers. Unlike Alexa, Google Assistant on a smartphone was capable of doing a few tasks locally on the device. For more complex actions, Google Assistant needed access to its cloud services.[59]

In 2025, Google began replacing Google Assistant with Gemini, its GenAI assistant. Gemini soon became the default assistant on Google's Pixel 9 phones. Users could access Gemini with voice or typing requests. Gemini was also one of the first GenAI-powered voice assistant technologies, giving it a head start over Apple Siri and Alexa. To ensure a smooth transition to Gemini, Google announced that it would end all access to Google Assistant in 2025. However, the transition from Assistant to Gemini was expected to be seamless for users, since both assistants were activated using the same trigger word (Hey Google). One estimate expected Google Assistant/Gemini users to increase from

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

24.2% of the U.S. population in 2022 to 27.7% in 2026.[60] Even though Gemini was growing, there was no model for monetization. (see **Exhibit 6**)

Pichai's goal was to have Gemini's voice assistant across devices, including cars, wrist devices/wearables, TVs, and in smart homes. In May 2025, Google announced it was adding its GenAI-powered Gemini into Android Auto for drivers to utilize the voice assistant features.[61] As one report stated, "drivers can speak naturally on [a] range of topics and get detailed responses without remembering specific phrases or interacting with the infotainment screen."[62] Google promoted it as a safe alternative for drivers to perform tasks and reduce distractions, stating that the voice assistant feature would be available in autos that supported Android Auto by the end of 2025. Additionally, Google announced a test pilot of GenAI-powered Gemini voice assistant software in Google Home. Conveniently named "Voice Assistant," the software's pilot assessed how multiple users in one home interacted via Voice Assistant with different actions available through Google's smart devices. Similar to Alexa+, Voice Assistant from Google recognized users' voices, and utilized this as a privacy layer by offering users two options for who could use Voice Assistant: Everyone (where any person could use the associated smart home devices), or Require Voice Match (which restricted use to only people who Voice Assistant recognized).[63] Google was also developing additional ways for users to interact with Google Home devices, including "triggers that are dependent on the weather and dates."[64]

Apple Apple, the consumer electronics company responsible for the Macintosh computer and the iPhone, was the first to market with a voice-activated intelligent assistant, Siri, launched in October 2011 with the iPhone 4S. Apple acquired Siri in 2010 from the research group SRI International. Apple maintained a closed ecosystem, developing its own applications and making Siri available only for Apple products until 2016. After Amazon's early success with Alexa, Apple opened its ecosystem to third-party applications. However, Apple restricted integration to specific applications, such as note-taking and ride-hailing, and in 2021, it stopped supporting many third parties entirely. Renowned for its strict stance on protecting consumer privacy, Apple tagged Siri searches with random user identifiers to protect consumer privacy and stored tagged data for only six months before deleting it. Observers noted that this decision reduced Apple's ability to train Siri and improve functionality.[65]

Apple had also released a line of smart speakers integrated with Siri. Known as HomePods, these smart speakers were introduced in February 2018. Initial reviews were mixed, with some critics stating that for the starting price of $349, the features were "minimal" while also being "over-engineered."[66] On the positive side, HomePod reportedly had better sound quality than most competitors. After failing to gain significant market share, Apple introduced the HomePod Mini ($99) in 2020.[67] In early 2023, Apple announced they were releasing the second generation of the HomePod, which was cheaper ($299), but with fewer features than the first-generation product.[68]

For years, Siri struggled compared to Google and Alexa.[69] Reviews noted that Siri often failed to answer basic questions correctly, such as "What month is it?".[70] To address this challenge, Apple announced Apple Intelligence in 2024, a new GenAI product included in iOS 18 and deployed on devices including the iPhone 16 and higher phone models. The plan was to integrate Apple Intelligence into Siri as a "gradual rollout," giving the company time to respond to any unexpected issues with the technology. Unfortunately, Apple struggled to deliver the promised advancements with Apple Intelligence and Siri in 2025.[71] One group of iPhone 16 owners sued Apple for false advertising, saying that many promised features had not been delivered, and Siri had only seen "a barebones…update."[72] For complex questions and tasks, Apple outsourced its GenAI service to OpenAI in June 2024.

In November 2024, Apple announced it was developing a major overhaul of Siri, powered by models developed by Apple. Named LLM Siri, it was explicitly designed to be used on smartphones and

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

enhance Siri's capabilities in handling more complex tasks. Additionally, LLM Siri would have an "expanded" ability to interact with third-party apps, and simultaneously use Apple Intelligence to summarize and compose text. According to Craig Federighi, LLM Siri would differ from ChatGPT:

> The properties of something like OpenAI advanced voice mode and Siri are quite different. That OpenAI mode is great if you want to go and ask a question about quantum mechanics and have it write a poem …It's not going to open your garage…There are many, many useful things Siri does for you every day, does them quickly and locally on your device.[73]

In March 2025, Apple announced that LLM Siri would be released "in the coming year."[74] Some analysts speculated that this delay in releasing the new version of Siri would cause Apple to fall further behind in the voice wars.[75] By July 2025, Apple leadership was getting discouraged about the company's ability to deliver Apple Intelligence with internal LLMs.[76] As a result, it had discussions with Google, Anthropic, and OpenAI about powering Siri. Anthropic reportedly asked for a multi-year, multi-billion-dollar annual commitment.[77]

OpenAI In November 2022, OpenAI released ChatGPT, a text-based interactive chatbot, which made OpenAI a household name. Although OpenAI released several versions over the next two years, all of them were text-based and did not include speech functions. In May 2024, ChatGPT released Advanced Voice Mode; in September, Advanced Voice Mode was available to all users, allowing people to communicate with the chatbot through speech.[78] Advanced Voice Mode could be interrupted by the user and was able to respond to emotions.[79] ChatGPT had live video and screen-sharing features, which enabled users to engage in real-time video chats, allowing ChatGPT to react to what it saw.[80]

Reviews were initially mixed. One report stated that the voice assistant's ability to help users with language translation and learning was excellent, with "one China analyst observed that using it to practice spoken Chinese has been 'nothing short of mind-blowing.'"[81] Others reported feeling "creepy… They are so lifelike and yet they are not 'right'… [a] void of strangeness."[82] After previewing Advanced Voice Mode in August 2024, one journalist stated:

> When counting quickly, it'll pretend it's out of breath. It says "um" and "uh." It laughs. It changes tone when talking about different or upbeat topics. After messing around with it, I felt equal parts amazed and uncomfortable. It sounds natural (when it isn't glitching), it responds in seconds, and you're able to interrupt it and ask new things – but do we need chatbots that sound like us?[83]

Another issue reported with OpenAI's voice assistant was accessibility. OpenAI's voice assistant could only be accessed on a computer or the company's smartphone app. When using the assistant, users noted an awkwardness due to unnatural lags and time delays between what the human user would speak to the assistant and how long Voice Mode took to respond, particularly on Android.[84]

In December 2024, OpenAI added a video feature to Advanced Voice Mode, letting the chatbot see through a phone camera and answer users' questions based on these images.[85] In March 2025, OpenAI updated Advanced Voice Mode to address a frequent complaint that Voice Mode would start speaking too quickly when users paused.[86] For paying users, the update also included "an improved personality" from Voice Mode.[87] An OpenAI spokesperson explained that the new voice assistant was "more direct, engaging, concise, specific, and creative in its answers."[88] In late August 2025, it upgraded its voice assistant again with its Realtime API, which connected AI models to apps, claiming to reduce latency and offer more natural conversations.[89] The primary market for this upgrade was

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

customer service chatbots. It lowered prices to $32/1M tokens for audio input and $64/1m audio output, which was still considered expensive by developers.[90]

OpenAI's Voice Mode also caused controversy. In May 2024, actress Scarlett Johansson threatened to sue OpenAI for copying her voice for ChatGPT Voice Mode.[91] The option, called "Sky," seemed to closely resemble Johansson's iconic voice, with her "clear, low tone and a hint of vocal fry."[92] Several Sky users commented on how the chatbot's voice sounded like Johansson's character in the movie *Her*, in which Johansson voiced a speaking chatbot character.[93] (Sam Altman had previously stated that *Her* was his favorite movie.)[94] OpenAI pulled the Sky option from Voice Mode to respect Johansson, and issued an apology.[95]

OpenAI was small relative to the other giant firms in the space, such as Apple, Google, Meta, and Amazon. Its estimated 2024 revenue was around $4 billion, with net losses of $5 billion.[96] However, OpenAI was growing rapidly, investing heavily in GenAI, and was well-funded.[97] (see **Exhibit 7**)

Meta Although Meta made early moves in the voice assistant space, its early devices were unsuccessful. In 2015, Meta launched a text-based virtual assistant called M. This voice assistant shut down in 2018.[98] In 2017, it acquired Ozlo, a conversational AI startup.[99] In February 2022, CEO Mark Zuckerberg said that Meta was developing a voice assistant, and that it needed to "advance beyond the current state of the art for smart assistants" to support his vision of the "metaverse," a digital space where people interacted with information and other human beings using VR and AR.[100]

While Amazon and Apple had been relatively slow in developing their own LLMs, Meta invested early and heavily in its own open-source LLMs, which could be leveraged for voice. In September 2024, Meta released LLAMA 3.2, a version of their open-source AI model that included voice capabilities for the first time.[101] LLAMA 3.2 included upgraded AI assistants that spoke with celebrities' voices, including Dame Judi Dench, Awkwafina, Kristen Bell, and John Cena.[102] Additionally, 3.2 had visual capabilities, allowing it to see the world around it through a device such as a smartphone camera, increasing its effectiveness in AI agents and similar roles. According to Meta, over 180 million people were already using its LLMs every week.[103]

One of Meta's most successful voice assistants was Meta Smart Glasses, developed in partnership with Ray Ban. In 18 months, Meta sold over 1 million pairs of Ray Ban glasses.[104] These eyeglasses were equipped with cameras, speakers, and microphones, allowing the wearer to engage with Meta's AI models and the internet without pulling out their phone. Additionally, users could take photos of something they were looking at in real time and use voice to ask Meta about the image. The AI model could perform searches using the image, providing answers in real-time. Meta AI also had an advantage over OpenAI's assistant because LLAMA could be downloaded and run locally without additional charges (except for commercial use, which had restrictions). LLAMA could also be fine-tuned more easily than OpenAI to specialize in specific tasks.[105]

In April 2025, Meta released a standalone app for Meta AI. This app was designed to compete with ChatGPT and Anthropic's Claude. The app gave "more personalized responses based on context and other user-specific elements related to the user's Facebook and Instagram accounts." It also paired with the Ray Ban smart glasses.[106] The Meta AI app, powered by LLAMA 4, was designed to compete with models from OpenAI, Google, Anthropic, and China's DeepSeek through better reasoning capabilities, multilingual capabilities, and increased efficiency.[107] The app focused on voice as the main mode of communication. Similar to OpenAI, it was designed to facilitate conversational interactions.[108]

China By 2025, demand for voice assistants in China had exploded, multiple options to consumers. Several companies utilized in-house AI models to power their voice assistants. As one Chinese analyst

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

noted, Chinese voice offerings were "deeply woven into the fabric of China's digital infrastructure, with features tailored to the nuances of Chinese culture, language, and commerce."[109] Baidu was one of the most popular voice assistants, providing a collection of smart speakers such as Xiaodu. Xiaomi offered the AI model XiaoAI, which operated in devices such as the Mi AI Speaker and was integrated across many of Xiaomi's products, particularly for smart home applications. Alibaba had its own smart speaker called Tmall Genie. The voice assistant, powered by AliGenie, offered voice-powered shopping and paying as an extension of their commerce platform.[110] Tencent's voice assistant, Xiaowei, held a smaller share of the Chinese voice assistant market. Designed to support services like WeChat and gaming platforms, Xiaowei served more as a tool to keep users within the Tencent ecosystem rather than a profit point for the company.[111] Similarly, ByteDance's Doubao, one of China's most popular consumer-facing AI apps, launched a real-time video call function that transformed the interface into an interactive digital assistant.[112] DeepSeek, the closest Chinese competitor to OpenAI, also released voice capabilities in its chatbot, allowing users to interact by voice, and integrating its voice assistant into other platforms such as Home Assistant and ElevenLabs.[113]

Not Just Smart Speakers or Smartphones: Other Devices

Voice assistants had a long history of integrating into devices other than smartphones and smart speakers. As GenAI increased voice assistants' functionalities, it made the paired devices more useful. Product companies began partnering with voice assistants, integrating the technology into consumer goods ranging from cars to wearables.

Cars Automobiles offered an obvious use case for voice assistants. A driver could easily do several things with a voice assistant, such as start a voice call or ask for directions, without taking their eyes off the road. Apple was the first to bring its voice assistant to cars. In 2014, several automakers introduced new vehicles featuring CarPlay, which allowed users to control their iPhones using the vehicle's touchscreen and summon Siri with a button on the steering wheel.[114] Later that year, Google unveiled Android Auto, which worked similar to CarPlay.[115] By 2025, most cars supported both.

As brands began integrating GenAI into their voice assistants, car companies began updating their voice assistants. In 2024, Amazon announced its partnership with BMW to introduce the BMW Intelligent Personal Assistant, powered by Amazon Alexa Custom Assistant. By January 2025, Amazon and BMW announced new LLM-based GenAI capabilities in the voice assistant as a test run in some vehicles and countries, which included the ability to plan trips and navigate more conversationally.

There appeared to be a strong demand for GenAI voice assistants in cars. According to an independent survey conducted by SoundHound in October 2024, 76% of U.S. drivers were "likely to use voice generative AI capabilities in their car if it was available," almost a 52% increase from 2023.[116] This increased to 82% of drivers between 35-44 years old, and 84% for people ages 21-34.[117] Additionally, 74% of those surveyed stated they would "be willing to pay a fee or be served ads in order to get access."[118] Some of the most desirable use cases for in-car voice assistants included in-vehicle food ordering (71% of respondents), navigation (85%), and texting or calling (76%).[119]

Wearables Wearable technology like smart watches and wireless earbuds were a natural platform for voice assistants. Siri was available on various earbuds and was accessible via the Apple Watch and Apple AirPods.[120] Similarly, Google's Pixel Buds featured easy access to Google Assistant.[121] In addition to smart glasses and wristwatches, other companies were creating wearables that contained voice assistants. In 2024, Apple shipped its Vision Pro, a spatial computer worn like ski goggles that could be partly controlled via voice.[122] In 2025, wearables with a GenAI-powered personal assistant

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

appeared in fitness tracker wristbands, which recorded and processed a user's conversations, and generated "personalized to-do lists and readable summaries of… in-person chats."[123]

Voice assistant devices varied in how they engaged with the user. Some companies focused on developing app ecosystems around their wearables to invite third-party apps to engage with users on their wearable GenAI voice-powered devices.[124] Unlike previous wearables, the computational requirements for GenAI meant that most devices processed voice commands in the cloud and a paired smartphone, rather than on separate hardware. Since GenAI-powered voice assistants were still relatively new, the verdict was out on consumer demand for these features.

Smart Home One of the most popular applications for voice assistants was smart homes. Consumers could turn on and off lights, appliances, adjust thermostats, etc. with voice commands. One of the biggest challenges for smart homes was a lack of standards. Alexa, Siri, Google, and others could control devices, but not talk with each other. In 2022, the industry realized that the lack of standards was inhibiting demand, and they worked together to create the "Matter 1.0" standard for smart home devices.[125] The Connectivity Standards Alliance (CSA) stated:

> This industry-unifying standard is a promise of reliable, secure connectivity... Matter creates more connections between more objects, simplifies development for manufacturers, and increases compatibility for consumers…By building upon Internet Protocol (IP), Matter will enable communication across smart home devices, mobile apps, and cloud services.[126]

Matter was an open-source protocol that could control all of a user's smart home devices without opening multiple apps or control centers; everything was centralized. Amazon, Google, and Apple supported the protocol. However, supporting the protocol did not always translate into full interoperability. As one report noted:

> While the big platform providers can see the benefit in a common standard, they are not going to open up full control of their devices to their competitors. There is a gap between the walled garden ecosystem experience and Matter functionality. So far, Matter functionality is limited, and manufacturers are keeping certain features proprietary.[127]

Challenges

Privacy, Hallucinations, and Glitches

One of the biggest concerns from voice assistant users was privacy. According to a 2025 survey, 73% of people stated "they have privacy concerns with AI integration" in voice assistants.[128] While voice assistants provided ease of use for many users, other people were concerned with how their data was being used and stored, and who exactly was listening. (see **Exhibits 8** and **9**) For example, Amazon stated that Echo devices were not constantly eavesdropping on users' conversations but instead would only listen when the trigger word "Alexa" was used in an always-listening microphone.[129] However, users began reporting issues with several brands' voice assistants, which would sometimes respond to conversations without a brand's trigger word. In 2018, one woman, who had implemented Alexa devices throughout her home, asked Amazon to check whether Alexa was maintaining privacy after Alexa recorded a conversation with her husband and sent it to one of her husband's coworkers, all without being prompted. After realizing the coworker heard everything in the conversation, the woman stated, "I felt invaded. A total privacy invasion. Immediately, I said, 'I'm never plugging that device in again because I can't trust it.'"[130] Other users reported that Alexa would start laughing in a

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

creepy manner without being prompted, and one stated that Alexa began listing local funeral homes and cemeteries.[131] Amazon faced increased backlash when it was revealed that the company allowed employees to listen to thousands of conversations recorded by Alexa devices. The company stated that these were being used to train devices and software. In response to the backlash, Amazon implemented new Echo privacy controls, which allowed people to opt out of sending their conversations to Amazon.[132] It also offered a way for users to delete video recordings from Alexa automatically, called "Auto Delete," which erased video recordings after three or 18 months, as well as an option that let users delete their location history.[133]

In 2021, a California resident filed a lawsuit that alleged that "Siri listened to private or confidential conversations" and "were then shared with third-party businesses…that then targeted…consumers with ads in Apple search and its Safari web browser." In 2025, Apple settled the class-action lawsuit for $95 million.[134] Similarly, in 2019, a Belgian journalist reported that he had been given over 1,000 files from a Google contractor who was paid by the company to review audio recorded by Google's smart speakers and other voice assistant devices. The files included information on people's addresses and private conversations among adults and children.[135]

With the introduction of GenAI, privacy became an even bigger challenge. LLMs relied heavily on user data to train and improve their models. In March 2025, Amazon announced that Echo users would not be able to opt out of using their data: Every command would be recorded and sent to AWS for processing.[136] In an email to Alexa users who opted into the original privacy feature, Amazon stated, "As we continue to expand Alexa's capabilities with generative AI features…, we have decided to no longer support this feature."[137]

Another concern surrounded the accuracy and relevancy of answers. One study conducted in 2024 found that Google Assistant was the most accurate, with 88% of its answers being correct. This was followed by Alexa (70%), Siri (52%), and ChatGPT 4 (35%). However, for relevant answers, the most accurate was GPT-4 (96%), followed by Google Assistant (92%), Alexa (82%), and Siri (76%). With the introduction of GenAI in voice assistants, there were additional concerns that GenAI voice assistants could make false or inaccurate statements and decisions. Known as hallucinations, these occurred when "a generative AI chatbot or computer vision tool perceives patterns or objects that are nonexistent or imperceptible to human observers, creating outputs that are nonsensical or altogether inaccurate."[138] Hallucinations often made it difficult to discern objective facts from false information. If a voice assistant was designed to be agentic and make decisions, the potential for negative consequences was serious. One example emerged in 2024, where OpenAI's automatic speech recognition system called Whisper began hallucinating. Although Whisper was designed to have almost a "human-level robustness and accuracy", one researcher at the University of Michigan reported hallucinations in 80% of the audio transcriptions he recorded and checked. Researchers were not able to identify the cause of these hallucinations. Still, they noted the hallucinations increased during "pauses, background sounds, or music playing," causing OpenAI to recommend that Whisper not be used in "decision-making contexts, where flaws in accuracy can lead to pronounced flaws in outcomes."[139]

Additional Costs Integrating GenAI into voice assistants added significant costs to traditional machine learning-based voice capabilities. Training costs for a sophisticated LLM could cost hundreds of millions of dollars.[140] In addition, the costs per query were high, especially considering that vendors were hoping for billions of interactions. OpenAI's cost per query was around $0.01 per 30 words, while Amazon Alexa+'s costs were estimated to be $0.02 per query.[141] Apple and Google did not disclose the cost of a Siri with Apple Intelligence or Gemini query, but their costs were likely similar.[142] Estimates varied, but GenAI queries could cost between 10 and 100 times more than a Google search query.[143]

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

Latency A common technical problem with integrating GenAI into voice assistants was latency, or a noticeable delay between the user's voice command and performing the task. According to a 2024 Amazon memo, "Latency remain[ed] a critical issue requiring significant improvements."[144]

Advertising in Voice Amazon and other companies tried to identify ways to make voice assistants profitable. One possibility was to incorporate paid advertising. However, it remained uncertain whether customers would accept a voice assistant that incorporated regular ads. Some users commented that they were frustrated with ad experiences on their voice assistants, with one user saying it "disgusts me a little…to think that I can't go without seeing ads on a device I purchased." They explained their frustration by stating, "Imagine getting a photo frame to display pictures but every 5th picture is an ad."[145]

In 2025, there was some evidence that consumer attitudes towards ads might be changing. Streaming media services, such as Netflix, initially offered subscriptions with no ads, but consumers became more open to advertising if it meant a lower price.[146] For voice services, researchers found that giving consumers more control in deciding how many advertisements they wanted to see affected whether people had negative feelings towards ads on voice assistants. Specifically, a combination of options for consumers, such as allowing users to select ads' topics, prevented many from having negative reactions to voice ads. As one researcher noted, "Clearly this was a situation where advertising is not going to be slipped in, like you do with a regular Google search or with Facebook feeds…But rather, you need to get users' buy-in."[147]

Jassy's Future for Voice

After a decade as the leader in smart speakers and one of the most successful innovators with voice technology, Andy Jassy had to confront the new competitive realities of GenAI-generated voice in 2025. On one hand, Alexa was a 'winner' in smart speakers and voice, with a huge installed base. Amazon also successfully built a platform for voice, with over 100,000 skills (even if the skills were not widely used). However, it never generated the profits or e-commerce revenues that Jeff Bezos envisioned. On the other hand, GenAI was changing the game. Capabilities were expanding, the number of competitors was growing, demand for voice was likely to grow, and voice would inevitably be embedded in a broad array of devices. Yet the potential for economic losses was greater than ever.

While Jassy was committed to moving forward with the introduction of Alexa+, there was no consensus on whether bundling Alexa+ with Amazon Prime would pay off financially. The Alexa team worried that charging for Alexa+ after a decade of free service would produce a consumer backlash. Jassy hoped that Alexa+ would lead to more customers signing up for Prime, but was that enough? Jassy also wondered if smart speakers would remain the dominant device in the home or elsewhere over the next decade. If voice capabilities were embedded into other devices, would Amazon become less relevant? If so, what should Amazon do? Advertising might be a promising revenue generator, but there was uncertainty about how consumers would respond to voice-based ads. Advertising was penetrating new arenas (such as streaming media) to keep prices low, and maybe Jassy should bet on consumers becoming more open to voice-based ads in the future? Finally, GenAI for voice was still in its infancy. To protect the bottom line and avoid the inevitable hallucinations, should Jassy try to be a leader or slow-roll its GenAI strategy, wait for the technology to improve, and costs to come down?

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

Exhibit 1 Summary of Financial Statements in Millions, 2019-2024

Amazon

	2019	2020	2021	2022	2023	2024
Revenue	280,522	386,064	469,822	513,983	574,785	637,959
COGS	165,536	233,307	272,344	288,831	304,739	326,288
R&D	35,931	42,740	56,052	73,213	85,622	88,544
SG&A	64,313	87,193	116,485	138,428	146,805	153,771
Operating income (loss)	14,541	22,899	24,879	13,348	36,852	68,593
Net income	11,588	21,331	33,364	(2,722)	30,425	59,248
EBITDA	36,330	48,079	59,312	55,269	85,515	120,468
Cash and cash equivalent	36,092	42,122	36,220	53,888	73,387	78,779
Total assets	225,248	321,195	420,549	462,675	527,854	624,894
Total liabilities	163,188	227,791	282,304	316,632	325,979	338,924
Total shareholders' equity	62,060	93,404	138,245	146,043	201,875	285,970
Gross margin	41%	40%	42%	44%	47%	49%
Return on equity	22%	27%	29%	(2%)	18%	24%
Market Cap at End of Year	798,809	999,962	1,668,073	1,604,280	1,059,464	1,784,656

Source: Amazon 10-Ks and Capital IQ

Google

	2019	2020	2021	2022	2023	2024
Revenue	161,857	182,527	257,637	282,836	307,394	350,018
COGS	71,896	84,732	110,939	126,203	133,332	146,306
R&D	26,018	27,573	31,562	39,500	45,427	47,530
SG&A	28,015	28,998	36,422	42,291	44,342	41,996
Operating income (loss)	35,928	41,224	78,714	74,842	84,293	114,186
Net income	34,343	40,269	76,033	59,972	73,795	100,118
EBITDA	47,570	54,903	88,987	88,317	96,239	129,497
Cash and cash equivalent	18,498	26,465	20,945	21,879	24,048	23,466
Total assets	275,909	319,616	359,268	365,264	402,392	450,256
Total liabilities	74,467	97,072	107,633	109,120	119,013	125,172
Total shareholders' equity	201,442	222,544	251,635	283,379	325,084	245,267
Gross margin	56%	54%	57%	55%	57%	58%
Return on equity	18%	19%	32%	24%	27%	33%
Market Cap at End of Year	798,711	993,976	1,391,632	1,956,706	1,344,543	1,751,504

Source: Google 10-Ks and Capital IQ.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

Apple

	2019	2020	2021	2022	2023	2024
Revenue	260,174	274,515	365,817	394,328	383,285	391,035
COGS	161,782	169,559	212,981	223,546	214,137	210,352
R&D	16,217	18,752	21,914	26,251	29,915	31,370
SG&A	18,245	19,916	21,973	25,094	24,932	26,097
Operating income (loss)	63,930	66,288	108,949	119,437	114,301	123,216
Net income	55,256	57,411	94,680	99,803	96,995	93,736
EBITDA	76,477	77,344	120,233	130,541	125,820	134,661
Cash and cash equivalent	48,844	38,016	34,940	23,646	29,965	29,943
Total assets	338,516	323,888	351,002	352,755	352,583	364,980
Total liabilities	248,028	258,549	287,912	302,083	290,437	308,030
Total shareholders' equity	90,488	65,339	63,090	50,672	62,146	56,950
Gross margin	38%	38%	42%	43%	44%	46%
Return on equity	56%	74%	147%	176%	172%	157%
Market Cap at End of Year	1,105,307	1,850,816	2,457,678	2,477,530	2,747,394	3,369,468

Source: Apple 10-Ks and Capital IQ

Meta

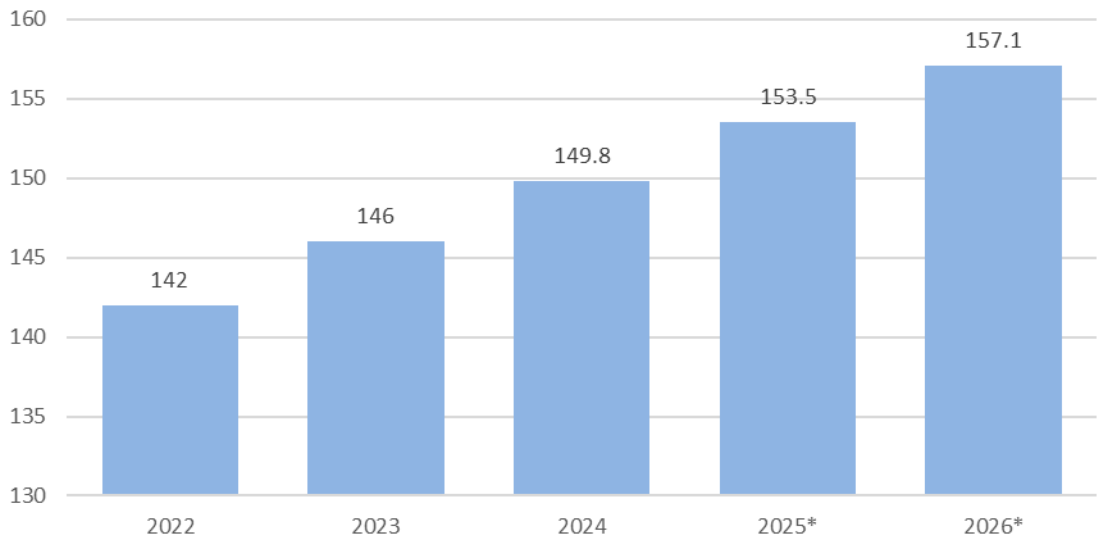
	2019	2020	2021	2022	2023	2024
Revenue	70,697	85,965	117,929	116,609	134,902	164,501
COGS	12,770	16,692	22,649	23,754	26,006	30,130
R&D	13,600	18,447	24,655	33,619	36,911	43,619
SG&A	15,341	18,155	23,872	25,681	22,979	22,533
Operating income (loss)	28,986	32,671	46,753	33,555	49,006	68,219
Net income	18,485	29,146	39,370	23,200	39,098	62,360
EBITDA	34,727	39,533	54,720	42,241	60,184	83,717
Cash and cash equivalent	19,079	17,576	16,601	14,681	41,862	43,889
Total assets	133,376	159,316	165,987	185,727	229,623	276,054
Total liabilities	32,322	31,026	41,108	60,014	76,455	93,417
Total shareholders' equity	101,054	128,290	124,879	125,713	153,168	182,637
Gross margin	82%	81%	81%	80%	82%	82%
Return on equity	20%	25%	31%	19%	28%	37%
Market Cap at End of Year	597,255	754,633	647,169	489,413	1,210,942	1,740,624

Source: Meta 10-Ks and Capital IQ.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

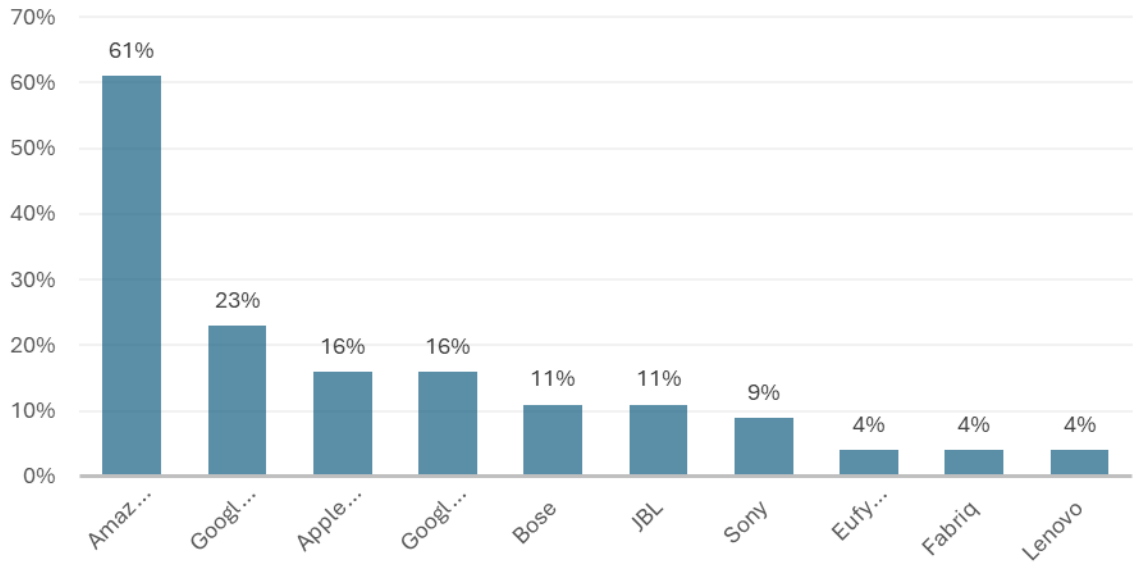
For the exclusive use of M. Bagadia, 2026.

Exhibit 2 Number of Voice Assistant Users in the United States, in millions



Source: Created by the casewriter using Statista and eMarketer, "Voice assistant users in the United States from 2022 to 2026 (in millions)." Statista, December 5, 2023, https://www.statista.com/statistics/1299985/voice-assistant-users-us/, accessed April 30, 2025. **Note**: 2025 and 2026 have asterisks because they are forecasts.

Exhibit 3 Percentage of People who Own Smart Speakers by Top 10 Brands in the U.S., Dec 2024

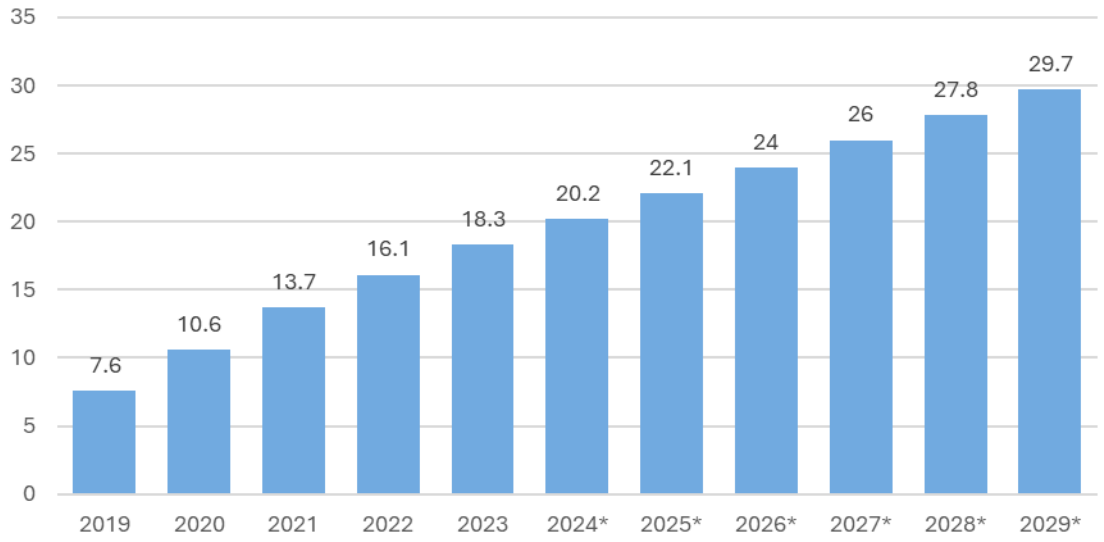


Source: Created by the casewriter using Statista Consumer Insights, "Smart speaker ownership by brand in the U.S. as of December 2024." January 30, 2025, https://www.statista.com/forecasts/997149/smart-speaker-ownership-by-brand-in-the-us, accessed April 30, 2025.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

Exhibit 4 Smart Speaker Market Revenue from 2019 to 2029 (in billions U.S. dollars)



Source: Created by the casewriter using Statista Consumer Insights, "Smart Speaker Market Revenue from 2019 to 2029 (in billions U.S. dollars)." June 26, 2025, https://www.statista.com/forecasts/1367973/smart-speaker-market-revenue-worldwide, accessed April 30, 2025. Note: 2024-2029 have asterisks because they are forecasts.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

Exhibit 5 Examples of U.S. Smart Speaker Retail Price in U.S. Dollars as of June 2025

Device	Price
Google Nest Mini (2nd Gen)	49
Google Nest Audio	99.99
Google Home (Original)	129
Amazon Echo Dot (5th Gen)	49.99
Amazon Echo (4th Gen)	99.99
Amazon Echo Studio	199.99
Apple HomePod Mini	99
Apple HomePod (2nd Gen)	299

Source: Created by the casewriter using data from Best Buy, Amazon, Apple, Google, and TechCrunch; https://store.google.com/config/google_nest_mini?hl=en-US&selections=eyJwcm9kdWN0RmFtaWx5IjoiWjI5dloyeGxYMjVsYzNSZmJXbHVhUT09In0%3D, https://www.bestbuy.com/site/google-nest-audio-smart-speaker-chalk/6428305.p?skuId=6428305&extStoreId=1126&utm_source=feed&ref=212&loc=19239471043&gad_source=1&gad_campaignid=19240877711&gbraid=0AAAAAD-ORIi_YSYfmSYeLW05LKL6Fj7zH&gclid=EAIaIQobChMI1snO-Z2XjgMV10b_AR0HMDVXEAQYASABEgKZvvD_BwE&gclsrc=aw.ds, https://techcrunch.com/2016/10/04/say-hello-to-google-home/, https://www.amazon.com/Amazon-vibrant-helpful-routines-Charcoal/dp/B09B8V1LZ3?th=1, https://www.nbcnews.com/select/shopping/new-amazon-echo-dot-ncna1241113, https://www.bestbuy.com/site/amazon-echo-studio-hi-res-smart-speaker-with-dolby-atmos-and-spatial-audio-processing-technology-and-alexa-glacier-white/6522274.p?skuId=6522274&extStoreId=536&utm_source=feed&ref=212&loc=19230189049&gad_source=1&gad_campaignid=19231438781&gbraid=0AAAAAD-ORIgv5P-hdQq3Cw0iFGAtbU8Mg&gclid=EAIaIQobChMIsd_11J6XjgMVMjYIBR2M7RigEAQYAiABEgIxwPD_BwE&gclsrc=aw.ds, https://www.apple.com/homepod-mini/, https://www.apple.com/homepod-2nd-generation/.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

Exhibit 6 Smartphone AI Features US Adults Currently Use or Plan to Use, by Age, Sep 2024

	18-34	35-54	55+	Total
Voice Assistant (e.g. searching the web, taking actions)	33%	28%	20%	26%
Photo and video tools (e.g. object removal, photo and video editing)	31%	25%	14%	22%
Advanced internet search engine	30%	20%	14%	20%
Visual search (e.g. image-based web search)	30%	21%	11%	20%
Writing tools (e.g. rewriting emails, summarizing)	30%	21%	11%	20%
Image or emoji generation	19%	15%	9%	14%
Organization tools (e.g. adding events mentioned in an email to your calendar)	17%	16%	9%	13%
Dictation and transcription tools	14%	11%	7%	10%
Other	0%	2%	2%	2%
Not applicable – I don't use or intend to use any smartphone AI-powered features	29%	44%	59%	45%

Source: Created by the casewriter using eMarketer and YouGov as cited in the company blog, "Smartphone AI Features US Adults Currently Use/Plan to Use, by Age, Sep 2024." September 30, 2025, https://www.emarketer.com/chart/268871/smartphone-ai-features-us-adults-currently-useplan-use-by-age-sep-2024-of-respondents-each-group, accessed April 30, 2025.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

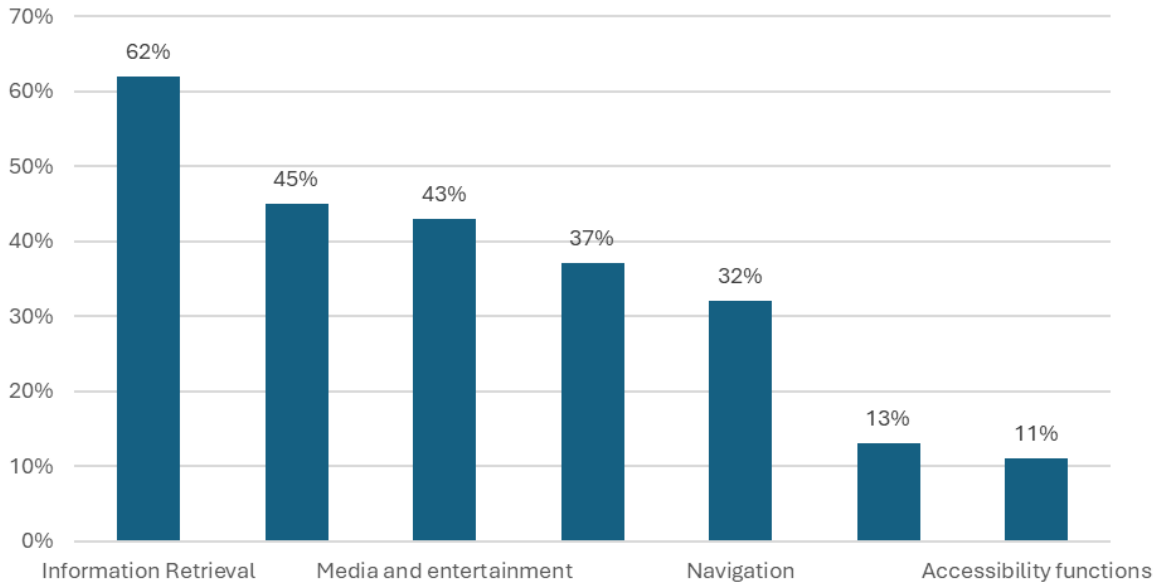
For the exclusive use of M. Bagadia, 2026.

Exhibit 7 OpenAI's Estimated Costs in 2024, in $USD

Category	Amount	Cumulative Amount
Total Revenue	$4B	$4B
Microsoft revenue share	($700M)	$3.3B
Compute to train models	($3B)	$0.3B
Compute to run models	($2B)	($1.7B)
Research compute amortization	($1B)	($2.7B)
Employee salaries	($700M)	($3.4B)
General & Admin	($600M)	($4B)
Data	($500M)	($4.5B)
Hosting, other cost of sales	($400M)	($4.9B)
Sales & Marketing	($300M)	($5.2B)
Loss, excluding stock-based compensation		$5.2 Billion

Source: Created by the casewriter using data from The Information: Cory Weinberg, "OpenAI Projections Imply Losses Tripling to $14 Billion in 2026." *The Information*, October 9, 2024, https://www.theinformation.com/articles/openai-projections-imply-losses-tripling-to-14-billion-in-2026?rc=pxkoyg, accessed June 24, 2025.

Exhibit 8 Percentage of US Voice Assistant Users asked about the Most Common Use Cases

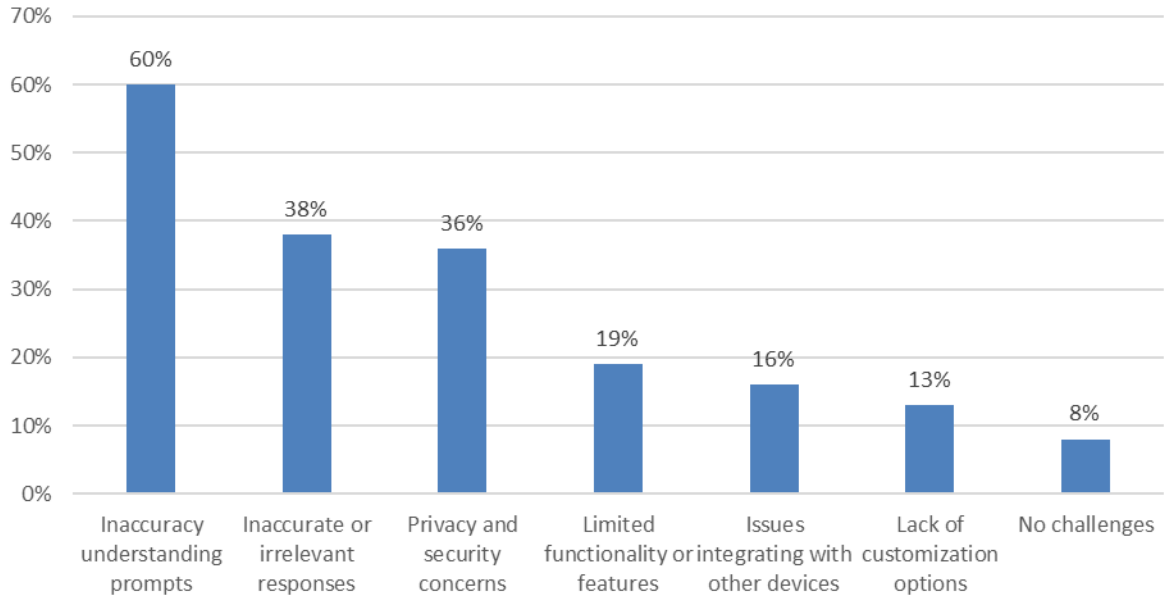


Source: Created by the casewriter using eMarketer & Voices, "Talking Technology: The Voice Assistant Said What?" conducted by Momentive, March 27, 2024, https://content-na1.emarketer.com/voice-assistant-user-forecast-2024, accessed April 20, 2025.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

Exhibit 9 Percentage of US Voice Assistant Users asked about their Concerns/Challenges using the Tech, Feb 2024



Source: Created by the casewriter using eMarketer & Voices, "Talking Technology: The Voice Assistant Said What?" conducted by Momentive, March 27, 2024, https://content-na1.emarketer.com/voice-assistant-user-forecast-2024, accessed April 20, 2025.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

Endnotes

[1] Jennifer Pattison Tuohy, "With Alexa Plus, Amazon finally reinvents its best product." *The Verge*, March 1, 2025, https://www.theverge.com/amazon-alexa/622172/amazon-alexa-plus-panos-panay-interview, accessed March 15, 2025.

[2] "Amazon CEO Jassy Talk Alexa+, AI Demand and Trump." *Bloomberg Television,* February 27, 2025, https://www.youtube.com/watch?v=1kGopJLJvgo, accessed April 13, 2025.

[3] Ron Amadeo, "Amazon Alexa is a 'colossal failure,' on pace to lose $10 billion this year." *Ars Technica*, November 21, 2022, https://arstechnica.com/gadgets/2022/11/amazon-alexa-is-a-colossal-failure-on-pace-to-lose-10-billion-this-year/, accessed May 1, 2025.

[4] Natalia Kuzminykh, "Voice Assistant Timeline: A Short History of the Voice Revolution." *Medium*, February 24, 2022, https://medium.com/geekculture/voice-assistant-timeline-50b3b07b48f7#:~:text=1950S%3A%20AUDREY,hands%2Dfree%20dialling%20or%20Audrey., accessed April 16, 2025.

[5] Natalia Kuzminykh, "Voice Assistant Timeline: A Short History of the Voice Revolution." *Medium*, February 24, 2022, https://medium.com/geekculture/voice-assistant-timeline-50b3b07b48f7#:~:text=1950S%3A%20AUDREY,hands%2Dfree%20dialling%20or%20Audrey., accessed April 16, 2025.

[6] Teddy Zhou, "How do voice assistants like Alexa & Siri actually work?" *Medium*, September 21, 2023, https://medium.com/@zhouxiaogan0/how-do-voice-assistants-like-alexa-siri-actually-work-1885dce1f683, accessed April 17, 2025.

[7] Teddy Zhou, "How do voice assistants like Alexa & Siri actually work?" *Medium*, September 21, 2023, https://medium.com/@zhouxiaogan0/how-do-voice-assistants-like-alexa-siri-actually-work-1885dce1f683, accessed April 17, 2025.

[8] Ahmed Sherif, "Which search do voice assistant users in the United States most use in 2022?" *UpCity and Statista*, November 3, 2023, https://www-statista-com/statistics/1384550/voice-assistant-search-united-states/, accessed April 14, 2025.

[9] Umair Bashir, "Smart speaker ownership by brand in the U.S. as of December 2024." *Statista*, January 2025, https://www-statista-com/forecasts/997149/smart-speaker-ownership-by-brand-in-the-us, accessed April 14, 2025.

[10] Jacob Borne, "Voice Assistant User Forecast 2024." *eMarketer*, August 21, 2024, https://emarketer.com/voice-assistant-user-forecast-2024, accessed April 15, 2025.

[11] Jacob Borne, "Voice Assistant User Forecast 2024." *eMarketer*, August 21, 2024, https://emarketer.com/voice-assistant-user-forecast-2024, accessed April 15, 2025.

[12] Jacob Borne, "Voice Assistant User Forecast 2024." *eMarketer*, August 21, 2024, https://content-na1.emarketer.com/voice-assistant-user-forecast-2024, accessed April 15, 2025.

[13] "Voice of a Generation: Why Gen-Z Prefers Voice Assistants." *Medium*, July 10, 2024, https://medium.com/@social_65128/voice-of-a-generation-why-gen-z-prefers-voice-assistants-2a105cacdd95, accessed May 27, 2025.

[14] Jacob Borne, "Voice Assistant User Forecast 2024." *eMarketer*, August 21, 2024, https://emarketer.com/voice-assistant-user-forecast-2024, accessed April 15, 2025.

[15] Jacob Borne, "Voice Assistant User Forecast 2024." *eMarketer*, August 21, 2024, https://emarketer.com/voice-assistant-user-forecast-2024, accessed April 15, 2025.

[16] Ayrin Ria, "A Brief History of Amazon." *Future Startup*, Oct 31, 2023, https://futurestartup.com/2023/10/31/a-brief-history-of-amazon/, accessed April 14, 2025.

[17] Brad Stone, "The Secret Origins of Amazon's Alexa." *WIRED*, May 11, 2021, https://www.wired.com/story/how-amazon-made-alexa-smarter/, accessed May 18, 2025.

[18] Brad Stone, "The Secret Origins of Amazon's Alexa." *WIRED*, May 11, 2021, https://www.wired.com/story/how-amazon-made-alexa-smarter/, accessed May 18, 2025.

[19] Brad Stone, "The Secret Origins of Amazon's Alexa." *WIRED*, May 11, 2021, https://www.wired.com/story/how-amazon-made-alexa-smarter/, accessed May 18, 2025.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

[20] Brad Stone, "The Secret Origins of Amazon's Alexa." *WIRED*, May 11, 2021, https://www.wired.com/story/how-amazon-made-alexa-smarter/, accessed May 18, 2025.

[21] Emma Bryce, "How Amazon's Alexa was 'born' and where voice-controlled tech will take us next."Wired (UK), February 14, 2017, https://www.wired.com/story/amazon-alexa-ai-evi/, accessed May 2, 2018.

[22] "The Amazon Echo Was Just the Beginning Of Alexa's Voice-First Revolution." *Forbes,* December 14, 2016, https://www.forbes.com/sites/quora/2016/12/14/the-amazon-echo-was-just-the-beginning-of-alexas-voice-first-revolution/, accessed May 2, 2018.

[23] Chris Welch, "Amazon just surprised everyone with a crazy speaker that talks to you." *The Verge*, November 6, 2014, available at: https://www.theverge.com/2014/11/6/7167793/amazon-echo-speaker-announced.

[24] Jide Edu, Xavier Ferrer-Aran, Jose Such, and Guillermo Suarez-Tangil, "Measuring Alexa Skill Privacy Practices across Three Years." In Proceedings of the ACM Web Conference 2022 (WWW '22), April 25–29, 2022, Virtual Event, Lyon, France. ACM, New York, NY, USA, 11 pages. https://doi.org/10.1145/ 3485447.3512289. Nilay Patel, "How Amazon runs Alexa, with Dave Limp." *The Verge*, October 12, 2021, available at: https://www.theverge.com/22719945/amazon-alexa-dave-limp-interview-ambient-computing-ring-eero-decoder.

[25] Khari Johnson, "Everything Amazon's Alexa learned to do in 2017." *VentureBeat,* December 29, 2017, available at: https://venturebeat.com/ai/everything-amazons-alexa-learned-to-do-in-2017/; Taylor Martin, "The 50 Best Alexa Skills That Help Make Your Life Much Easier." *CNET*, March 18, 2022, available at: https://www.cnet.com/home/smart-home/the-50-best-alexa-skills-that-help-make-your-life-so-much-easier/; Grant Clauser and Signe Brewster, "Amazon Alexa vs. Google Assistant: Which Voice Assistant Is Best for You?" *New York Times*, September 25, 2020, available at: https://www.nytimes.com/wirecutter/reviews/amazon-echo-vs-google-home/.

[26] Maxwell Zaff, "Amazon CEO says 100,000 users now have Alexa+." *TechCrunch*, May 1, 2025, https://techcrunch.com/2025/05/01/amazon-ceo-says-100000-users-now-have-alexa/, accessed May 5, 2025.

[27] Jacob Borne, "Voice Assistant User Forecast 2024." *eMarketer*, August 21, 2024, https://content-na1.emarketer.com/voice-assistant-user-forecast-2024, accessed April 15, 2025.

[28] Dan Alaimo, "The Information: Only 2% of Alexa users reportedly purchase off the devices." *RetailDive*, August 8, 2018, https://www.retaildive.com/news/the-information-only-2-of-alexa-users-reportedly-purchase-off-the-devices/529608/, accessed May 8, 2025.

[29] Wilson Lau, "Voice Search and Advertising: Adapting to Changing Consumer Behaviors." *AdRoll*, April 2024, https://www.adroll.com/blog/voice-search-and-advertising-adapting-to-changing-consumer-behaviors, accessed May 27, 2025.

[30] "How Voice-activated Search is Reshaping Video Advertising." *MTC Martechcube*, February 24, 2025, https://www.martechcube.com/voice-activated-search-is-reshaping-video-advertising/, accessed May 28, 2025.

[31] Jon Porter, "Andy Jassy officially takes over as Amazon CEO from Jeff Bezos." *The Verge*, July 5, 2021, https://www.theverge.com/2021/7/5/22563671/andy-jassy-amazon-ceo-jeff-bezos, accessed April 17, 2025.

[32] Subrat Patnaik and Jeran Wittenstein, "Amazon Becomes World's First Public Company to Lose $1 Trillion in Market Value." *Bloomberg*, November 9, 2022, https://www.bloomberg.com/news/articles/2022-11-09/amazon-hits-unwelcome-milestone-with-1-trillion-in-value-lost, accessed May 1, 2025.

[33] Ron Amadeo, "Amazon Alexa is a 'colossal failure,' on pace to lose $10 billion this year." *Ars Technica*, November 21, 2022, https://arstechnica.com/gadgets/2022/11/amazon-alexa-is-a-colossal-failure-on-pace-to-lose-10-billion-this-year/, accessed May 1, 2025.

[34] Eugene Kim, "Amazon's amazing turnaround: How Andy Jassy secretly wooed Wall Street and overhauled a bloated tech giant." *Business Insider*, July 8, 2024, https://www.businessinsider.com/amazon-andy-jassy-wooed-wall-street-overhauled-bloated-tech-giant-2024-7, accessed April 21, 2025.

[35] Eugene Kim, "Amazon's amazing turnaround: How Andy Jassy secretly wooed Wall Street and overhauled a bloated tech giant." *Business Insider*, July 8, 2024, https://www.businessinsider.com/amazon-andy-jassy-wooed-wall-street-overhauled-bloated-tech-giant-2024-7, accessed April 21, 2025.

[36] Eugene Kim, "Amazon's amazing turnaround: How Andy Jassy secretly wooed Wall Street and overhauled a bloated tech giant." *Business Insider*, July 8, 2024, https://www.businessinsider.com/amazon-andy-jassy-wooed-wall-street-overhauled-bloated-tech-giant-2024-7, accessed April 21, 2025.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

[37] Eugene Kim, "Amazon's amazing turnaround: How Andy Jassy secretly wooed Wall Street and overhauled a bloated tech giant." *Business Insider*, July 8, 2024, https://www.businessinsider.com/amazon-andy-jassy-wooed-wall-street-overhauled-bloated-tech-giant-2024-7, accessed April 21, 2025.

[38] Eugene Kim, "Amazon's amazing turnaround: How Andy Jassy secretly wooed Wall Street and overhauled a bloated tech giant." *Business Insider*, July 8, 2024, https://www.businessinsider.com/amazon-andy-jassy-wooed-wall-street-overhauled-bloated-tech-giant-2024-7, accessed April 21, 2025.

[39] Jennifer Pattison Tuohy, "With Alexa Plus, Amazon finally reinvents its best product." *The Verge*, March 1, 2025, https://www.theverge.com/amazon-alexa/622172/amazon-alexa-plus-panos-panay-interview, accessed March 15, 2025.

[40] Jennifer Pattison Tuohy, "With Alexa Plus, Amazon finally reinvents its best product." *The Verge*, March 1, 2025, https://www.theverge.com/amazon-alexa/622172/amazon-alexa-plus-panos-panay-interview, accessed March 15, 2025.

[41] "The Amazon Prime Membership Fee." *Amazon Help and Customer Service*, https://www.amazon.com/gp/help/customer/display.html?nodeId=G34EUPKVMYFW8N2U, accessed May 18, 2025.

[42] Mark Wilson, "Alexa+ explained 9 things you need to know about Amazon's new AI-powered assistant." *TechRadar*, February 26, 2025, https://www.techradar.com/home/smart-home/alexa-plus-price-release-date-features-everything-you-need-to-know, accessed May 5, 2025.

[43] "Amazon Nova Foundation Models." *AWS*, https://aws.amazon.com/ai/generative-ai/nova/, accessed May 5, 2025.

[44] Maxwell Zeff, "Amazon unveils Nova Act, an AI agent that can control a web browser." *TechCrunch*, March 31, 2025, https://techcrunch.com/2025/03/31/amazon-unveils-nova-act-an-ai-agent-that-uses-a-web-browser/, accessed May 3, 2025.

[45] Kathryn Rush, "I Got Early Access to Amazon's New Gen AI Alexa+. Things got Weird." *The New York Times,* June 12, 2025, https://www.nytimes.com/wirecutter/reviews/alexa-plus-gen-ai-preview/, accessed June 24, 2025.

[46] Jennifer Pattison Tuohy, "With Alexa Plus, Amazon finally reinvents its best product." *The Verge*, March 1, 2025, https://www.theverge.com/amazon-alexa/622172/amazon-alexa-plus-panos-panay-interview, accessed March 15, 2025.

[47] Teaganne Finn and Amanda Downie, "Agentic AI vs. generative AI." *IBM*, February 11, 2025, https://www.ibm.com/think/topics/agentic-ai-vs-generative-ai, accessed April 22, 2025.

[48] Kathryn Rush, "I Got Early Access to Amazon's New Gen AI Alexa+. Things got Weird." *The New York Times,* June 12, 2025, https://www.nytimes.com/wirecutter/reviews/alexa-plus-gen-ai-preview/, accessed June 24, 2025.

[49] Kevin Roose, "Alexa Got an A.I. Brain Transplant. How Smart is it Now?" New York Times, August 9, 2025. https://www.nytimes.com/2025/08/09/business/alexa-artificial-intelligence-amazon.html?smid=em-share, accessed August 12, 2025.

[50] Ibid.

[51] Ibid.

[52] Anna Fleck, "Alexa, What's America's Favorite Smart Speaker?" *Statista Consumer Insights*, March 28, 2025, https://www.statista.com/chart/23943/share-of-us-adults-who-own-smart-speakers/, accessed April 22, 2205.

[53] "Sundar Pichai, CEO, Google and Alphabet." *Congress.gov*, December 2019, https://www.congress.gov/116/meeting/house/110883/witnesses/HHRG-116-JU05-Bio-PichaiS-20200729.pdf, accessed March 27, 2025.

[54] Sundar Pichai, "A personal Google, just for you." *Google The KeywordI, October 4, 2016,* https://blog.google/products/assistant/personal-google-just-you/, accessed March 29, 2025.

[55] Janko Rottgers, "Google Officially Introduces Google Home, a $129 Smart, Wi-Fi Powered Speaker." *Variety, October 4, 2016,* https://variety.com/2016/digital/news/google-home-introduction-1201877695/, accessed April 11, 2025.

[56] S Pangambam, "Google I/O 2016 Keynote Full Transcript." *The Singju Post*, May 20, 2016, https://singjupost.com/google-io-2016-keynote-full-transcript/, accessed June 24, 2025.

[57] Jacob Borne, "Voice Assistant User Forecast 2024." *eMarketer*, August 21, 2024, https://emarketer.com/voice-assistant-user-forecast-2024, accessed April 15, 2025.

[58] Paul Thurrott, "Google Assistant Now Supports Over 1 Million Actions." *Thurrott*, January 9, 2018,

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

https://www.thurrott.com/smart-home/149900/google-assistant-now-supports-1-million-actions, accessed April 28, 2025.

[59] "Local fulfillment." Google: *Cloud-to-cloud Developer Center*, https://developers.home.google.com/local-home/overview#:~:text=Local%20Execution%20provides%20the%20ability,making%20a%20cloud%20API%20call., accessed June 24, 2025.

[60] Lisa Haiss, "The end is nigh for genAI-free Google Assistant." *eMarketer*, March 17, 2025, https://emarketer.com/end-nigh-genai-google-assistant, accessed April 28, 2025.

[61] Eric Walz, "Google is adding its AI-powered Gemini voice assistant to Android Auto." *Automotive Dive*, May 19, 2025, https://www.automotivedive.com/news/google-adds-gemini-AI-androidd-auto-automotive-os-voice-assistant/748120/, accessed June 24, 2025.

[62] Eric Walz, "Google is adding its AI-powered Gemini voice assistant to Android Auto." *Automotive Dive*, May 19, 2025, https://www.automotivedive.com/news/google-adds-gemini-AI-androidd-auto-automotive-os-voice-assistant/748120/, accessed June 24, 2025.

[63] Nickolas Diaz, "Google Home kicks off a Gemini-backed 'Voice Assistant' test for Android." *Android Central*, May 26, 2025, https://www.androidcentral.com/accessories/smart-home/google-home-gemini-voice-assistant-test-android, accessed June 24, 2025.

[64] Nickolas Diaz, "Google Home kicks off a Gemini-backed 'Voice Assistant' test for Android." *Android Central*, May 26, 2025, https://www.androidcentral.com/accessories/smart-home/google-home-gemini-voice-assistant-test-android, accessed June 24, 2025.

[65] Tripp Mickle, "I'm Not Sure I Understand'- How Apple's Siri Lost Her Mojo." *Wall Street Journal,* June 7, 2017, https://www.wsj.com/articles/apples-siri-once-an-original-now-struggles-to-be-heard-above-the-crowd-1496849095, accessed April 10, 2018.

[66] "Encore: The Return of the HomePod." *BasicAppleGuy*, Feb 2023, https://basicappleguy.com/basicappleblog/encorethereturnofthehomepod, accessed May 18, 2025.

[67] Samuel Axon, "Apple pulls the plug on the HomePod." *Ars Technica*, March 15, 2021, https://arstechnica.com/gadgets/2021/03/apple-pulls-the-plug-on-the-homepod/, accessed May 18, 2025. Jon Porter, "Apple announces revamped full-size HomePod two years after discontinuing original." *The Verge*, January 18, 2023, https://www.theverge.com/2023/1/18/23553366/apple-homepod-2022-price-release-date-specs-features, accessed May 18, 2025.

[68] Jon Porter, "Apple announces revamped full-size HomePod two years after discontinuing original." *The Verge*, January 18, 2023, https://www.theverge.com/2023/1/18/23553366/apple-homepod-2022-price-release-date-specs-features, accessed May 18, 2025.

[69] Filipe Esposito, "While we worry about Apple Intelligence, Siri can't answer 'what month is it?'"

9to5Mac, March 19, 2025, https://9to5mac.com/2025/03/19/siri-cant-answer-what-month-is-it/, accessed April 6, 2025.

[70] Filipe Esposito, "While we worry about Apple Intelligence, Siri can't answer 'what month is it?'"

9to5Mac, March 19, 2025, https://9to5mac.com/2025/03/19/siri-cant-answer-what-month-is-it/, accessed April 6, 2025.

[71] Mark Gurman and Drake Bennett, "Why Apple Still Hasn't Cracked AI." *Bloomberg*, May 18, 2025, https://www.bloomberg.com/news/features/2025-05-18/how-apple-intelligence-and-siri-ai-went-so-wrong, accessed May 21, 2025.

[72] Emily Forlini, "Alexa+ Takes a Page From Apple Intelligence With Staggered AI Rollout." *PC Mag*, April 1, 2025 https://www.pcmag.com/news/alexa-plus-takes-a-page-from-apple-intelligence-with-staggered-ai-rollout, accessed April 7, 2025.

[73] Emma Roth, "Apple is reportedly working on 'LLM Siri' to compete with ChatGPT." *The Verge*, November 21, 2024, https://www.theverge.com/2024/11/21/24302685/apple-llm-siri-compete-chatgpt, accessed April 28, 2025.

[74] Grace Harmon, "Delays for an upgraded Siri could cost Apple in voice assistant race." *eMarketer*, March 7, 2025, https://emarketer.com/delays-upgraded-siri-could-cost-apple-voice-assistant-race, accessed April 28, 2025.

[75] Grace Harmon, "Delays for an upgraded Siri could cost Apple in voice assistant race." *eMarketer*, March 7, 2025, https://emarketer.com/delays-upgraded-siri-could-cost-apple-voice-assistant-race, accessed April 28, 2025.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

[76] Mark Gurman, "Apple Weighs Using Anthropic or OpenAI to Power Siri in Major Reversal, July 1, 2025, https://www.bloomberg.com/news/articles/2025-06-30/apple-weighs-replacing-siri-s-ai-llms-with-anthropic-claude-or-openai-chatgpt?fromMostRead=true&sref=E9Urfma4&mc_cid=13ce23f388&mc_eid=001d905869, accessed July 1, 2025.

[77] Ibid.

[78] Maxwell Zeff, "OpenAI rolls out Advanced Voice Mode with more voices and a new look." *TechCrunch*, September 24, 2024, https://techcrunch.com/2024/09/24/openai-rolls-out-advanced-voice-mode-with-more-voices-and-a-new-look/, accessed April 2, 2025.

[79] Sabrina Ortiz, "OpenAI's voice mode rolling out to all ChatGPT subscribers – here's who gets it first." *ZDNET*, September 24, 2024, https://www.zdnet.com/article/openais-voice-mode-rolling-out-to-all-chatgpt-subscribers-heres-who-gets-it-first/, accessed April 29, 2025.

[80] Gadjo Sevilla, "AI's live video and screen-sharing agents could redefine mobile shopping's future." *eMarketer*, March 3, 2025, https://emarketer.com/ai-s-live-video-screen-sharing-agents-could-redefine-mobile-shopping-s-future, accessed April 28, 2025.

[81] Catherine Thorbecke, "ChatGPT is now a creepy cultural chameleon." *The Japan Times*, December 4, 2024, https://www.japantimes.co.jp/commentary/2024/12/04/world/chatgpt-creepy-cultural-chameleon/, accessed April 29, 2025.

[82] Catherine Thorbecke, "ChatGPT is now a creepy cultural chameleon." *The Japan Times*, December 4, 2024, https://www.japantimes.co.jp/commentary/2024/12/04/world/chatgpt-creepy-cultural-chameleon/, accessed April 28, 2025; Rina Diane Caballar, "What Is the Uncanny Valley?" *IEEE Spectrum*, November 11, 2024, https://spectrum.ieee.org/what-is-the-uncanny-valley, accessed April 28, 2025.

[83] Kylie Robinson, "OpenAI's new voice mode threw me into the uncanny valley." *The Verge*, August 15, 2024, https://www.theverge.com/2024/8/15/24220378/openai-advanced-voice-mode-uncanny-valley, accessed April 29, 2025.

[84] "Severe Lag in Advanced Voice and Video Modes in Android – Persistent Issue Since Launch." *OpenAI Developer Community*, Dec 2024, https://community.openai.com/t/severe-lag-in-advanced-voice-and-video-modes-on-android-persistent-issue-since-launch/1060824, accessed May 26, 2025.

[85] Cecily Mauran, "OpenAI brings video to ChatGPT Advanced Mode." *Mashable*, December 12, 2024, https://mashable.com/article/openai-brings-video-to-chatgpt-advanced-voice-mode, accessed April 2, 2025.

[86] Ananya Gairola, "OpenAI Says It's Making ChatGPT Voice Assistant Sound More Like A Human And It Won't Interrupt As Much." *Benzinga*, March 24, 2025, https://www.benzinga.com/tech/25/03/44468019/openai-says-its-making-chatgpt-voice-assistant-sound-more-like-a-human-and-it-wont-interrupt-as-much, accessed April 29, 2025.

[87] Maxwell Zeff, "OpenAI says its AI voice assistant is now better to chat with." *TechCrunch*, March 24, 2025, https://techcrunch.com/2025/03/24/openai-says-its-ai-voice-assistant-is-now-better-to-chat-with/, accessed April 29, 2025.

[88] Maxwell Zeff, "OpenAI says its AI voice assistant is now better to chat with." *TechCrunch*, March 24, 2025, https://techcrunch.com/2025/03/24/openai-says-its-ai-voice-assistant-is-now-better-to-chat-with/, accessed April 29, 2025.

[89] https://openai.com/index/introducing-gpt-realtime/, accessed September 2, 2025.

[90] Mervin Praison, "GPT Reatime Voice App: I Replaced Custmer Service with 2 AI Agents, https://openai.com/index/introducing-gpt-realtime/, accessed September 2, 2025.

[91] Hannah Gearan, "Scarlett Johansson Takes Legal Action Against OpenAI For Seemingly Copying Her Voice." *ScreenRant*, May 21, 2204, https://screenrant.com/scarlett-johansson-voice-ai-lawsuit-imitation/, accessed May 14, 2025.

[92] Tripp Mickle, "Scarlett Johansson Said No, but OpenAI's Virtual Assistant Sounds Just Like Her." *New York Times*, May 20, 2024, https://www.nytimes.com/2024/05/20/technology/scarlett-johannson-openai-voice.html, accessed May 14, 2025.

[93] Hannah Gearan, "Scarlett Johansson Takes Legal Action Against OpenAI For Seemingly Copying Her Voice." *ScreenRant*, May 21, 2204, https://screenrant.com/scarlett-johansson-voice-ai-lawsuit-imitation/, accessed May 14, 2025.

[94] Tripp Mickle, "Scarlett Johansson Said No, but OpenAI's Virtual Assistant Sounds Just Like Her." *New York Times*, May 20, 2024, https://www.nytimes.com/2024/05/20/technology/scarlett-johannson-openai-voice.html, accessed May 14, 2025.

[95] Hannah Gearan, "Scarlett Johansson Takes Legal Action Against OpenAI For Seemingly copying her Voice." *ScreenRant*, May 21, 2204, https://screenrant.com/scarlett-johansson-voice-ai-lawsuit-imitation/, accessed May 14, 2025.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

[96] Shef Wang, "AI Doesn't Print Money, Business Does." *The Artificial Intelli'science*, December 2, 2024, https://www.intelli-science.com/p/ai-doesnt-print-money-business-does, accessed May 25, 2025.

[97] Hayden Field and Kate Rooney, "OpenAI closes $40 billion funding round, largest private tech deal on record." *CNBC*, April 1, 2025, https://www.cnbc.com/2025/03/31/openai-closes-40-billion-in-funding-the-largest-private-fundraise-in-history-softbank-chatgpt.html#:~:text=OpenAI%20closes%20$40%20billion%20funding%20round%2C%20largest,$10%20billion%20from%20a%20syndicate%20of%20investors, accessed June 6, 2025.

[98] Casey Newton, "Facebook is shutting down M, its personal assistant service that combined humans and AI." *The Verge*, January 8, 2018, available at: https://www.theverge.com/2018/1/8/16856654/facebook-m-shutdown-bots-ai.

[99] John Mannes, "Facebook buys Ozlo to boost its conversational AI efforts." *TechCrunch*, July 31, 2017, available at https://techcrunch.com/2017/07/31/facebook-buys-ozlo-to-boost-its-conversational-ai-efforts/.

[100] Shirin Ghaffrary, "Mark Zuckerberg wants to build a voice assistant that blows Alexa and Siri away." *Vox*, February 23, 2022, available at: https://www.vox.com/recode/22948097/mark-zuckerberg-voice-assistant-metaverse-ai-announcement.

[101] Reuters, "Facebook parent Meta Platforms launches standalone AI assistant app." *Reuters*, April 29, 2025, https://www.reuters.com/business/facebook-parent-meta-platforms-launches-standalone-ai-assistant-app-2025-04-29/?utm_source=chatgpt.com, accessed April 29, 2025.

[102] Will Knight, "Meta Releases Llama 3.2 – and Gives Its AI a Voice." *Wired*, September 25, 2024, https://www.wired.com/story/meta-releases-new-llama-model-ai-voice/, accessed April 30, 2025.

[103] Will Knight, "Meta Releases Llama 3.2 – and Gives Its AI a Voice." *Wired*, September 25, 2024, https://www.wired.com/story/meta-releases-new-llama-model-ai-voice/, accessed April 30, 2025.

[104] Sherin Shibu, "Meta is Reportedly Working on Smart Glasses With a Screen – at a Price Point $1,000 More Than Its Ray-Bans." *Entrepreneur*, April 2, 2025, https://www.entrepreneur.com/business-news/meta-is-reportedly-working-on-smart-glasses-with-a-screen/489460, accessed April 7, 2025.

[105] Will Knight, "Meta Releases Llama 3.2 – and Gives Its AI a Voice." *Wired*, September 25, 2024, https://www.wired.com/story/meta-releases-new-llama-model-ai-voice/, accessed April 30, 2025.

[106] Alex Heath, "Meta's ChatGPT competitor shows how your friends use AI." *The Verge*, April 29, 2025, https://www.theverge.com/ai-artificial-intelligence/657645/meta-ai-app-chatgpt-competitor-release-ios-android?utm_source=chatgpt.com, accessed April 29, 2025.

[107] Reuters, "Facebook parent Meta Platforms launches standalone AI assistant app." *Reuters*, April 29, 2025, https://www.reuters.com/business/facebook-parent-meta-platforms-launches-standalone-ai-assistant-app-2025-04-29/?utm_source=chatgpt.com, accessed April 29, 2025.

[108] Alex Heath, "Meta's ChatGPT competitor shows how your friends use AI." *The Verge*, April 29, 2025, https://www.theverge.com/ai-artificial-intelligence/657645/meta-ai-app-chatgpt-competitor-release-ios-android?utm_source=chatgpt.com, accessed April 29, 2025.

[109] Akihabara News, "Chinese Smart Speakers." *Akihabara News*, December 24, 2024, https://akihabaranews.com/chinese-smart-speakers/?utm_source=chatgpt.com, accessed April 30, 2025.

[110] Akihabara News, "Chinese Smart Speakers." *Akihabara News*, December 24, 2024, https://akihabaranews.com/chinese-smart-speakers/?utm_source=chatgpt.com, accessed April 30, 2025

[111] Akihabara News, "Chinese Smart Speakers." *Akihabara News*, December 24, 2024, https://akihabaranews.com/chinese-smart-speakers/?utm_source=chatgpt.com, accessed April 30, 2025

[112] Coco Feng, "ByteDance Upgrades Doubao AI App with Real-Time Interactive Video Call Function," *South China Morning Post*, May 27, 2025. https://www.scmp.com/tech/tech-trends/article/3311978/bytedance-upgrades-doubao-ai-app-real-time-interactive-video-call-function, accessed July 22, 2025.

[113] Kelly Ng, Brandon Drennon, Tom Gerken, and Marc Cieslak, "DeepSeek: The Chinese AI app that has the world talking." *BBC*, Feb 4, 2025, https://www.bbc.com/news/articles/c5yv5976z9po, accessed June 6, 2025.; "Conversation agents, HA vs Deepseek (or others)." *Home Assistant Configuration page*, January 28, 2025, https://community.home-assistant.io/t/conversation-agents-ha-vs-deepseek-or-others/837788?utm_source=chatgpt.com, accessed June 6, 2025.; Amanda Caswell, "I just chatted with DeepSeek – here's how to try it yoursef with ElevenLabs' ne voice integration." *Tom's*

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

Guide, January 31, 2025, https://www.tomsguide.com/ai/i-just-chatted-with-deepseek-heres-how-to-try-it-yourself-with-elevenlabs-new-voice-integration?utm_source=chatgpt.com, accessed June 6, 2025.

[114] Katie Holliday, "Had the iCar arrived? Apple launches in-car platform." CNBC, March 3, 2014, https://www.cnbc.com/2014/03/03/has-the-icar-arrived-apple-launches-in-car-app.html, accessed May 6, 2025.

[115] Kif Leswing, "Apple's massive success with CarPlay paves the way for automotive ambitions." *CNBC*, May 29, 2021, https://www.cnbc.com/2021/05/29/apple-carplay-massive-success-paves-way-for-automotive-entry.html, accessed May 6, 2025.

[116] "Research Finds 76% Of U.S. Drivers Likely To Use Voice Generative AI Capabilities In Their Vehicles." *BusinessWire*, October 22, 2024, https://www.businesswire.com/news/home/20241022788245/en/Research-Finds-76-Of-U.S.-Drivers-Likely-To-Use-Voice-Generative-AI-Capabilities-In-Their-Vehicles, accessed May 7, 2025.

[117] "Research Finds 76% Of U.S. Drivers Likely To Use Voice Generative AI Capabilities In Their Vehicles." *BusinessWire*, October 22, 2024, https://www.businesswire.com/news/home/20241022788245/en/Research-Finds-76-Of-U.S.-Drivers-Likely-To-Use-Voice-Generative-AI-Capabilities-In-Their-Vehicles, accessed May 7, 2025.

[118] "Research Finds 76% Of U.S. Drivers Likely To Use Voice Generative AI Capabilities In Their Vehicles." *BusinessWire*, October 22, 2024, https://www.businesswire.com/news/home/20241022788245/en/Research-Finds-76-Of-U.S.-Drivers-Likely-To-Use-Voice-Generative-AI-Capabilities-In-Their-Vehicles, accessed May 7, 2025.

[119] "Research Finds 76% Of U.S. Drivers Likely To Use Voice Generative AI Capabilities In Their Vehicles." *BusinessWire*, October 22, 2024, https://www.businesswire.com/news/home/20241022788245/en/Research-Finds-76-Of-U.S.-Drivers-Likely-To-Use-Voice-Generative-AI-Capabilities-In-Their-Vehicles, accessed May 7, 2025.

[120] David Pogue, "Ingenious New Earbuds From Arriva." *New York Times*, May 31, 2012, via Factiva.

[121] Ashley Carman, "Google's Pixel Buds are the company's first wireless headphones." *The Verge*, April 1, 2021, https://www.theverge.com/circuitbreaker/2017/10/4/16401132/google-pixel-buds-wireless-headphones-announced-price-release-date, accessed May 7, 2025.

[122] "Apple Vision Pro available in the U.S. on February 2." *Apple*, January 8, 2024, https://www.apple.com/newsroom/2024/01/apple-vision-pro-available-in-the-us-on-february-2/, accessed May 18, 2025. "Apple Vision Pro User Guide." *Apple*, https://support.apple.com/guide/apple-vision-pro/perform-actions-with-your-voice-tan14d179ad1/visionos, accessed May 18, 2025.

[123] Julian Chokkattu, "Your Next AI Wearable Will Listen to Everything All the Time." *WIRED*, January 8, 2025, https://www.wired.com/story/bee-ai-omi-always-listening-ai-wearables/, accessed May 19, 2025.

[124] Julian Chokkattu, "Your Next AI Wearable Will Listen to Everything All the Time." *WIRED*, January 8, 2025, https://www.wired.com/story/bee-ai-omi-always-listening-ai-wearables/, accessed May 19, 2025.

[125] "Matter: The Foundation of Connected Things." *Connectivity Standards Alliance*, https://csa-iot.org/all-solutions/matter/, accessed May 8, 2025.

[126] "Matter: The Foundation of Connected Things." *Connectivity Standards Alliance*, https://csa-iot.org/all-solutions/matter/, accessed May 8, 2025.

[127] Simon Hill, "Here's What the 'Matter' Smart Home Standard Is All About." *WIRED*, June 9, 2024, https://www.wired.com/story/what-is-matter/, accessed May 8, 2025.

[128] Tyler Lacoma, "Amazon's Alexa AI Upgrade Remains in Early Access CNET Survey Shows 73% of Device Users Express Privacy Worries." *CNET*, May 14, 2025, https://www.cnet.com/home/smart-home/amazons-alexa-ai-upgrade-remains-in-early-access-cnet-survey-shows-73-of-device-users-express-privacy-worries/, accessed May 26, 2025.

[129] Sam Wolfson, "Amazon's Alexa recorded private conversation and sent it to random contact." *The Guardian*, May 24, 2018, https://www.theguardian.com/technology/2018/may/24/amazon-alexa-recorded-conversation, accessed April 15, 2025.; "Alexa laughing at users." *Fox10 News*, March 7, 2018, https://www.youtube.com/watch?v=z7t9f_x258Q, accessed April 15, 2025.

[130] Sam Wolfson, "Amazon's Alexa recorded private conversation and sent it to random contact." *The Guardian*, May 24, 2018, https://www.theguardian.com/technology/2018/may/24/amazon-alexa-recorded-conversation, accessed April 15, 2025.

[131] "Alexa laughing at users." *Fox10 News*, March 7, 2018, https://www.youtube.com/watch?v=z7t9f_x258Q, accessed April 15, 2025.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.

For the exclusive use of M. Bagadia, 2026.

[132] Scharon Harding, "Everything You Say to Your Echo Will Soon Be Sent to Amazon, and You Can't Opt Out." *WIRED*, March 17, 2025, https://www.wired.com/story/everything-you-say-to-your-echo-will-be-sent-to-amazon-starting-march-28/, accessed April 15, 2025.

[133] Christine Fisher, "Amazon adds Auto Delete for Alexa following privacy concerns." *Engadget*, September 25, 2019, https://www.engadget.com/2019-09-25-amazon-alexa-auto-delete-video-recordings.html, accessed April 15, 2025.

[134] Aimee Picchi, "Apple applying $95 million in a Siri eavesdropping settlement. Here's how to file a claim." *CBS News*, May 14, 2025, https://www.cbsnews.com/news/apple-siri-settlement-95-million-lopez-how-to-file-claim/, accessed May 26, 2025.

[135] Tom Simonite, "Who's Listening When You Talk to Your Google Assistant?" *WIRED*, July 10, 2019, https://www.wired.com/story/whos-listening-talk-google-assistant/, accessed May 26, 2025.

[136] Scharon Harding, "Everything You Say to Your Echo Will Soon Be Sent to Amazon, and You Can't Opt Out." *WIRED*, March 17, 2025, https://www.wired.com/story/everything-you-say-to-your-echo-will-be-sent-to-amazon-starting-march-28/, accessed April 15, 2025.

[137] Scharon Harding, "Everything You Say to Your Echo Will Soon Be Sent to Amazon, and You Can't Opt Out." *WIRED*, March 17, 2025, https://www.wired.com/story/everything-you-say-to-your-echo-will-be-sent-to-amazon-starting-march-28/, accessed April 15, 2025.

[138] Garance Burke and Hilke Schellmann, "Experts warn of AI's 'hallucinations' in medical, legal transcriptions." *Katv*, October 26, 2024, https://katv.com/news/nation-world/researchers-say-an-ai-powered-transcription-tool-used-in-hospitals-invents-things-no-one-e-fabrication-transcripts-artificial-intelligence-white-house-alondra-nelson-princeton, accessed May 26, 2025.

[139] Garance Burke and Hilke Schellmann, "Experts warn of AI's 'hallucinations' in medical, legal transcriptions." *Katv*, October 26, 2024, https://katv.com/news/nation-world/researchers-say-an-ai-powered-transcription-tool-used-in-hospitals-invents-things-no-one-e-fabrication-transcripts-artificial-intelligence-white-house-alondra-nelson-princeton, accessed May 26, 2025.

[140] Dorothy Neufeld, "Visualizing the Training Costs of AI Models Over Time." *Visual Capitalist*, June 4. 2024, https://www.visualcapitalist.com/training-costs-of-ai-models-over-time/?utm_source=chatgpt.com, accessed June 23, 2025.

[141] Rahul Makwana, "Apple's Innovative Approach to ChatGPT Integration in Siri." *LinkedIn*, June 16, 2024, https://www.linkedin.com/pulse/apples-innovative-approach-chatgpt-integration-siri-rahul-makwana-8xqkc?utm_source=chatgpt.com, accessed June 23, 2025; Maxwell Zeff, "Alexa Will Get an Upgrade But It'll Cost You, Report Says." *Gizmodo*, May 22, 2024, https://gizmodo.com/amazon-alexa-generative-ai-upgrade-prime-cost-1851493383?utm_source=chatgpt.com, accessed June 23, 2025.

[142] Rahul Makwana, "Apple's Innovative Approach to ChatGPT Integration in Siri." *LinkedIn*, June 16, 2024, https://www.linkedin.com/pulse/apples-innovative-approach-chatgpt-integration-siri-rahul-makwana-8xqkc?utm_source=chatgpt.com, accessed June 23, 2025.

[143] https://www.nextbigfuture.com/2024/04/if-ai-llm-queries-replace-google-internet-search.html, accessed July 22, 2025.

[144] Jason Del Rey, "Exclusive: Leaked Amazon documents identify critical flaws in the delayed AI reboot of Alexa." *Yahoo!Tech and Fortune*, December 6, 2024, https://tech.yahoo.com/ai/articles/exclusive-leaked-amazon-documents-identify-005500770.html, accessed May 18, 2025.

[145] "Infuriated by Ads on Echo Show home content." *Reddit*, https://www.reddit.com/r/amazonecho/comments/1dwp3y9/infuriated_by_ads_on_echo_show_home_content/, accessed May 27, 2025.

[146] Marisa Jones, "Most consumers are receptive to ads on streaming services," May 21, 2025. https://www.emarketer.com/content/most-consumers-receptive-ads-on-streaming-services, accessed July 1, 2025.

[147] "More user control may help ease negative reactions to ads on voice assistants." *Penn State*, October 24, 2024, https://www.psu.edu/news/bellisario-college-communications/story/more-user-control-may-help-ease-negative-reactions-ads, accessed May 28, 2025.

This document is authorized for use only by Mihir Bagadia in ISM 6485 - 2026 taught by Emre Demirezen, University of Florida from Mar 2026 to Aug 2026.